Math 1231 Assignment

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Question 1

The function $T : \mathbb{R}^2 \to \mathbb{R}^3$ is defined by

$$T\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ 5x_1 + 5x_2 \\ x_2 \end{pmatrix} \text{ for all } \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2.$$

Show that T is linear.

Solution. First we know the domain and codomain of T are in known vector spaces namely $\mathbb{R}^2$ and $\mathbb{R}^3$ respectively. So to see if it's linear we need to check addition and scalar multiplication conditions.

Addition condition. For $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^2$, we have $\mathbf{x} + \mathbf{x}' = \begin{pmatrix} x_1 + x_1' \\ x_2 + x_2' \end{pmatrix} \in \mathbb{R}^2$, and hence

$$\begin{aligned} T(\mathbf{x} + \mathbf{x}') &= \begin{pmatrix} x_1 + x_1' \\ 5(x_1 + x_1') + 5(x_2 + x_2') \\ x_2 + x_2' \end{pmatrix} \\ &= \begin{pmatrix} x_1 \\ 5x_1 + 5x_2 \\ x_2 \end{pmatrix} + \begin{pmatrix} x_1' \\ 5x_1' + 5x_2' \\ x_2' \end{pmatrix} \\ &= T(\mathbf{x}) + T(\mathbf{x}'). \end{aligned}$$

Thus, the addition condition is satisfied.

Scalar multiplication condition. For $\mathbf{x} \in \mathbb{R}^2$ and $\lambda \in \mathbb{R}$, we have $\lambda\mathbf{x} = \begin{pmatrix} \lambda x_1 \\ \lambda x_2 \end{pmatrix} \in \mathbb{R}^2$, and hence

$$T(\lambda\mathbf{x}) = \begin{pmatrix} \lambda x_1 \\ \lambda(5x_1) + \lambda(5x_2) \\ \lambda x_2 \end{pmatrix} = \lambda \begin{pmatrix} x_1 \\ 5x_1 + 5x_2 \\ x_2 \end{pmatrix} = \lambda T(\mathbf{x}).$$

Thus, the scalar multiplication condition is also satisfied, and therefore T is a linear map.

Question 2

Show that

$$S = \{\mathbf{x} \in \mathbb{R}^4 : 3x_1 - 4x_2 - 8x_3 - 2x_4 = 0\}$$

is a subspace of $\mathbb{R}^4$.

Solution. First to figure out if S is a subspace of $\mathbb{R}^4$, we need to apply the Subspace Theorem. This states that a subset S of a vector space V over a field $\mathbb{F}$, under the same rules for addition and multiplication by scalars, is a subspace of V if and only if

i) the vector $\mathbf{0}$ in V also belongs to S.

ii) S is closed under addition

iii) S is closed under scalar multiplication by scalars from $\mathbb{F}$.

Check for the vector 0. This is simply done as $3(0) - 4(0) - 8(0) - 2(0) = 0$, so the zero vector of $\mathbb{R}^4$ is in S.

Closure under addition. For any vector $\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} \in S$ we have

$$3u_1 - 4u_2 - 8u_3 - 2u_4 = 0 \tag{1}$$
$$3v_1 - 4v_2 - 8v_3 - 2v_4 = 0. \tag{2}$$

If we add (1) and (2), we have

$$(3u_1 - 4u_2 - 8u_3 - 2u_4) + (3v_1 - 4v_2 - 8v_3 - 2v_4) = 0 + 0.$$

Hence, we obtain

$$3(u_1 + v_1) - 4(u_2 + v_2) - 8(u_3 + v_3) - 2(u_4 + v_4) = 0,$$

and so $\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \\ u_4 + v_4 \end{pmatrix} \in S$. Thus, S is closed under addition.

Closure under scalar multiplication. To check this, let $\lambda \in \mathbb{R}$ be a scalar. Then we can take equation (1) and multiply both sides by said λ. This gives

$$\lambda\,(3u_1 - 4u_2 - 8u_3 - 2u_4) = \lambda \cdot 0$$
$$\implies 3(\lambda u_1) - 4(\lambda u_2) - 8(\lambda u_3) - 2(\lambda u_4) = 0.$$

Hence, $\lambda\mathbf{u} = \begin{pmatrix} \lambda u_1 \\ \lambda u_2 \\ \lambda u_3 \\ \lambda u_4 \end{pmatrix} \in S$ and so S is closed under scalar multiplication.

By the Subspace Theorem, the set S is a vector subspace of $\mathbb{R}^4$.

Question 3

Let $\mathbb{P}_n$ be the set of real polynomials of degree at most n, and write p' and p'' for the first and second derivatives of p. Show that

$$S = \{p \in \mathbb{P}_6 : p''(-4) + 8p'(-4) = 0\}$$

is a subspace of $\mathbb{P}_6$.

Solution. Similar to question 2, to figure out if S is a subspace of $\mathbb{P}_6$, we use the Subspace Theorem. But, I will use the Alternative Subspace Theorem for this example. This theorem states A subset S of a vector space V over a scalar field $\mathbb{F}$ is a subspace of V if and only if S contains the zero vector and satisfies the closure condition:

$$\text{If } \mathbf{v}_1, \mathbf{v}_2 \in S \text{ and } \lambda_1, \lambda_2 \in \mathbb{F}, \text{ then } \lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 \in S.$$

Check for the vector 0. This is clearly in set S as $p''(-4) + 8p'(-4) = 0 + 8(0)$, so the zero polynomial of $\mathbb{P}_6$ is in S.

Check closure condition. Suppose that p, $q \in S$ and $\lambda_1, \lambda_2 \in \mathbb{F}$, then for $\lambda_1 p + \lambda_2 q$, we have

$$\begin{aligned}
(\lambda_1 p + \lambda_2 q)''(-4) + 8(\lambda_1 p + \lambda_2 q)'(-4) &= [\lambda_1 p''(-4) + \lambda_2 q''(-4)] + 8[\lambda_1 p'(-4) + \lambda_2 q'(-4)] \\
&= [\lambda_1 p''(-4) + \lambda_1 8p'(-4)] + [\lambda_2 q''(-4) + \lambda_2 8q'(-4)] \\
&= \lambda_1[p''(-4) + 8p'(-4)] + \lambda_2[q''(-4) + 8q'(-4)] \\
&= (\lambda_1 \cdot 0 + \lambda_1 \cdot 0) + (\lambda_2 \cdot 0 + \lambda_2 \cdot 0) \\
&= 0.
\end{aligned}$$

Hence, $\lambda_1 p + \lambda_2 q \in S$, and S is a subspace.

Note

$$p(x) = a_0 + a_1(x+4) + a_2(x+4)^2 + ... + a_6(x+4)^6.$$

Differentiating, we get

$$\begin{aligned}
p'(x) &= a_1 + 2a_2(x+4) + ... + 6a_6(x+4)^5 \\
&\implies p'(-4) = a_1 \\
p''(x) &= 2a_2 + 6a_3(x+4) + ... + 30a_6(x+4)^4 \\
&\implies p''(-4) = 2a_2.
\end{aligned}$$

We are told $p''(-4) + 8p'(-4) = 0$ then

$$\begin{aligned}
2a_2 &= -8a_1 \\
a_2 &= -4a_1.
\end{aligned}$$

This means that a basis of S is in the form

$$1, (x+4) - 4(x+4)^2, (x+4)^3, (x+4)^4, (x+4)^5, (x+4)^6,$$

because $a_1(x+4) + a_2(x+4)^2$ is replaced by $a_1((x+4) - 4(x+4)^2)$. Also, because each basis vector has an arbitrary real coefficient, $\dim S = 6$.

Question 4

The air in a 82 cubic metre kitchen is initially clean, but when Margaret burns her toast while making breakfast, smoke is mixed with the room's air at a rate of 0.08 mg per second. An air conditioning system exchanges the mixture of air and smoke with clean air at a rate of 9 cubic metres per minute. Assume that the pollutant is mixed uniformly throughout the room and that burnt toast is taken outside after 75 seconds. Let $S(t)$ be the amount of smoke in mg in the room at time t (in seconds) after the toast first began to burn.

a) Find a differential equation obeyed by $S(t)$.

b) Find $S(t)$ for $0 \leq t \leq 75$ by solving the differential equation in (a) with an appropriate initial condition.

c) What is the level of pollution in mg per cubic metre after 75 seconds?

d) How long does it take for the level of pollution to fall to 0.003 mg per cubic metre after the toast is taken outside?

Solution. a) First we need to find the differential equation for $\frac{dS(t)}{dt}$. First we know that $S(0) = 0$, and we also know how the smoke in the room S changes with time t. In particular,

$$\begin{aligned} \frac{dS(t)}{dt} &= \text{rate of change of } S \\ &= \text{(rate of inflow) - (rate of outflow).} \end{aligned} \tag{1}$$

Now we know that rate of inflow of smoke S is 0.08 mg per second. To calculate the rate of outflow, we take the total volume of the air in the kitchen at time t given by

$$82 - 9(t) + 9(t) = 82.$$

Hence, because the volume of air in the kitchen $V(t)$ does not change with time t the proportion of smoke is

$$\frac{S(t)}{82}.$$

Since the outflow of the air is 9 cubic metres per minute

$$\frac{S(t)}{82} \times \frac{9}{60}\ m^3/s = \frac{0.15}{82} S(t).$$

Following from (1), we obtain the initial value problem while the toast is burning $0 \leq t \leq 75$,

$$\frac{dS}{dt} = 0.08 - \frac{0.15}{82} S(t), \qquad S(0) = 0.$$

b) We solve the IVP (Initial Value Problem) found in a), say $k = \frac{0.15}{82}$ then,

$$\frac{dS}{dt} + k\, S = 0.08 \tag{1}$$

$$h(t) = e^{\int k \, \mathrm{d}t} = e^{kt}. \tag{2}$$

Multiplying the ODE (1) by e^{kt} (2) gives

$$\begin{aligned} e^{kt}\frac{dS}{dt} + e^{kt}k\,S &= e^{kt}0.08 \\ \frac{d}{dt}(e^{kt}S(t)) &= e^{kt}0.08 \\ e^{kt}S(t) &= \int (0.08e^{kt}dt) \\ e^{kt}S(t) &= \frac{0.08}{k}e^{kt} + C \\ S(t) &= \frac{0.08}{k} + Ce^{-kt}. \end{aligned}$$

Since $S(0) = 0$ and substituting k back into the equation, we have

$$\begin{aligned} \frac{656}{15} + Ce^{-\frac{0.15}{82}\times 0} &= 0 \\ C &= -\frac{656}{15}. \end{aligned}$$

So the amount of smoke in the kitchen at a particular time is given by

$$\begin{aligned} S(t) &= \frac{656}{15} - \frac{656}{15}e^{-\frac{0.15}{82}t} \\ &= \frac{656}{15}(1 - e^{-\frac{0.15}{82}t}). \end{aligned} \tag{3}$$

c) Using equation (3), when $t = 75$,

$$\begin{aligned} S(75) &= \frac{656}{15}(1 - e^{-\frac{0.15}{82}(75)}) \\ &\approx 5.61 \quad 3s.f. \end{aligned}$$

This is the amount of smoke (mg) in the kitchen. To get it in mg per cubic metre we need to divide by $V = 82$, we then have

$$\text{concentration} = \frac{5.61}{82} \approx 0.0684 \quad 3s.f.$$

d) For the level of pollution to fall to 0.003 mg per cubic metre, this can be figured out using a similar process as before. The differential equation is the same as equation (1) without the inflow like

$$\frac{dS(t)}{dt} = -\frac{0.15}{82}S(t).$$

Now like before say $k = \frac{0.15}{82}$ solving the IVP,

$$\begin{aligned} \int \frac{dS(t)}{dt} &= -\int \left(\frac{0.15}{82}\,S(t)\right) dt. \implies \\ S(t) &= e^{-kt+C} \\ &= Ce^{-kt}. \end{aligned}$$

Since $S(75) = 5.61$ then

$$S(75) = Ce^{-k75} \implies C = S(75)e^{k75}.$$

Substituting it back giving time taken after the toast is taken outside, we have

$$S(t) = S(75)e^{-k(t-75)}, \quad t \geq 75. \tag{4}$$

With equation (4) we can set the equation for $S(t) = 0.003 \times 82$ mg per cubic metre giving

$$S(75)e^{-\frac{0.15}{82}(t-75)} = 0.246, \quad t \approx 1709 \text{ seconds}.$$